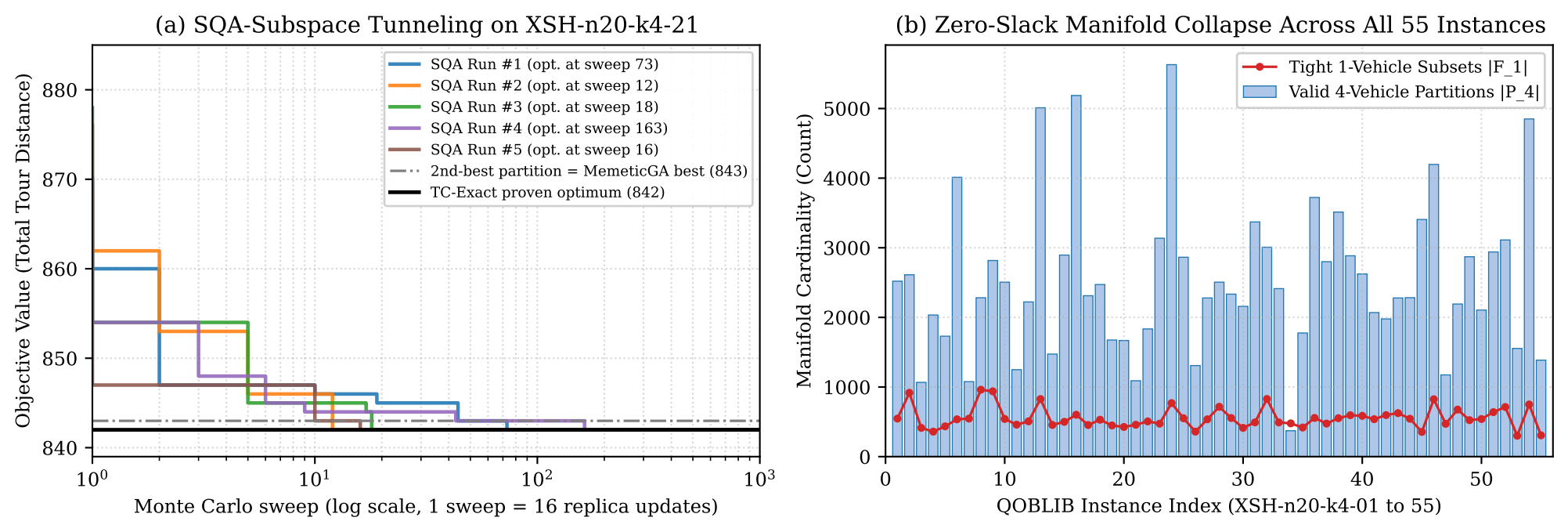
(a) SQA-Subspace Tunneling on XSH-n20-k4-21
Objective Value (Total Tour Distance)
Monte Carlo sweep (log scale, 1 sweep = 16 replica updates)
SQA Run #1 (opt. at sweep 73)
SQA Run #2 (opt. at sweep 12)
SQA Run #3 (opt. at sweep 18)
SQA Run #4 (opt. at sweep 163)
SQA Run #5 (opt. at sweep 16)
2nd-best partition = MemeticGA best (843)
TC-Exact proven optimum (842)
880
870
860
850
840
10^0
10^1
10^2
10^3
(b) Zero-Slack Manifold Collapse Across All 55 Instances
Manifold Cardinality (Count)
QOBLIB Instance Index (XSH-n20-k4-01 to 55)
Tight 1-Vehicle Subsets |F_1|
Valid 4-Vehicle Partitions |P_4|
5000
4000
3000
2000
1000
0
0
10
20
30
40
50